



Service Bulletin Number: 5659748

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PSBU Oil Approval and Oil Drain Extension Process Overview

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This service bulletin defines the methods a customer should follow to extend their oil drain interval without making modifications to the engine, for example, increasing the oil pan size. If the customer is interested in increasing the service life of other components such as filter change interval, refer to documents such as Service Bulletin 4388840 and 5411272. Ensure lubricating oil monitoring system is in place as specified in Section 4 of Service Bulletins 4388840 and 5411272.

Oil Certification Process

Issue:

- Customer is interested in having an oil approved that is not on the Cummins List of Approved Oils or extending oil drain interval for a Cummins Approved Oil through Cummins or on their own.

If the customer is interested in having an oil that is not on the Cummins Oil Registration List approved onto the Cummins Oil Registration List, the customer should submit an Initiation Request to their local distributor to begin the oil approval process. The distributor should contact Corporate Chemical Tech. Engineering (CCT) with the customer's request to begin the oil analysis trial process. The oil company will be involved in the approval process as numerous tests will need to be completed to understand if the oil meets the Cummins Standards. If the oil completes the certification process, the oil will be added to the Cummins Oil Registration list, referenced in the Fluids for Cummins Products Service Manual – 00-378-004 Determining Proper Engine Oil for Cummins Engines,

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html.

If the customer is interested in extending the oil drain interval beyond the agreed guidelines specified in the Owner's manual, they can follow one of the following processes:

- **Cummins Approved Oil – CMI Recognized Oil Drain Interval Extension Process**

If the customer is interested in having the oil drain interval extension formally recognized by Cummins, the customer should advise the local distributor and the distributor should contact CCT at fluids.registration@cummins.com to begin the trial. There is a cost associated with having this oil drain interval extension formally recognized by Cummins. If the customer chooses to participate in this process it will ensure the continuity of warranty.

- **Cummins Approved Oil – Customer Led Oil Drain Interval Extension Process**

If the customer is not interested in having the oil drain interval formally recognized by Cummins, the customer will need to contact the local distributor or Cummins Field Service Engineer / Divisional Field Service Engineer (CFSE/DFSE). The local distributor will provide the customer with relevant oil approval documentation and the customer will use that to identify the oil field test requirements. The documentation of the lubricating oil drain interval validation must be retained in the site assessment file. Failure to provide this information will void the warranty. The customer should provide details on the engine model, application, and site conditions. Along with these details the following should be considered: lubricating oil system capacity (including any remote lubricating oil tanks where the lubricating oil is continuously circulated), use of Centinel™ lubricating oil replenishment systems, use of Eliminator™ versus spin on lubricating oil filters and centrifuge, duty cycle, and lubricating oil used (brand name, performance category, and viscosity grade). If the fleet is made up of mixed engine models, varying duty cycles, or equipment with differing lubricating oil system configurations, each be considered separately. The oil field test process and requirements are different for diesel and gas engines.

The customer should determine the expected drain interval based off the oil analysis results and ECM data at hand. The engine must be new or rebuilt having 1500 hours or less. There must be a minimum of 2 engines and 3 drains per engine. For gas engines, if multiple engines are not available, a minimum of 3 drains per engine will suffice. Oil samples are to be taken every 250 hours or a minimum of 5 samples per drain, whichever is shorter. The customer will begin the field test. They are expected to send oil samples to oil analyses labs that can meet the requirements in the table below, gather engine ECM data and determine the duty cycle. Oil samples must be taken from the oil filter head upstream of the filters. It is recommended to follow the engine oil sampling procedure found in the 5411406 Fluids for Cummins Products Service Manual. Refer to 00-378-011 Engine Oil Sampling Procedure using the oil sampling valve.

When collecting oil samples, it is important to label the samples with the following information before submitting for analysis:

- Engine model/serial number
- Hours of lubricating oil use
- Hours accumulated on engine since new or rebuild
- Lubricating oil used (brand name, performance category, and viscosity grade)
- Date sample was collected
- Engine application
- Amount of new lubricating oil added since previous lubricating oil change

Half way through the field test a mid-project review (after first set of engines goes through 1st oil drain interval) must be conducted to review the oil sample results to determine the performance of the oil. The oil sample data should also be compared with historic sample data from the previous oil used. During the review, it will be determined whether the oil meets the used oil criteria per the Cummins Fluid Manual. Refer to 00-378-004 Determining Proper Engine Oil for Cummins Engines. Refer to 00-378-005 Diesel Engine Oil Requirements and 00-378-006 Natural Gas and Liquified Petroleum Gas Engine Oil Requirements for the respective requirements. For diesel engines, the oil analysis must meet the criteria found in Service Bulletin 4022060: Oil Analysis Techniques for High Horsepower Diesel Engines. If any of the oil analyses parameters exceed the oil caution or condemn limits, then the oil trial must be stopped. If the oil meets the criteria, the field test can continue, and the same steps should be followed to gather engine ECM

data, determine duty cycle and send oil samples to the oil labs until the end of the field test. It is best practice, to use the same oil analysis lab for the duration of the trial to obtain to most consistent and accurate results. At the end of the field test, oil samples analysis results should be reviewed to determine the extended oil drain interval. The filter change interval does not change. It should be determined whether the oil meets the used oil criteria per the Cummins Fluid Manual. Refer to 00-378-004 Determining Proper Engine Oil for Cummins Engines. Refer to 00-378-005 Diesel Engine Oil Requirements and 00-378-006 Natural Gas and Liquified Petroleum Gas Engine Oil Requirements for the respective requirements. For diesel engines, the oil analysis must meet the criteria found in Service Bulletin 4022060: Oil Analysis Techniques for High Horsepower Diesel Engines. If the criteria are not met, the request should be rejected by the customer. If the criteria are met, the customer should document the data.

For Gas engines, refer to Service Manual 4021313 Section V 66-018-003 for the Lubricating Oil recommendations and Specifications.

For diesel engines, the recommended ASTM Methods are below for each oil parameter.

Used Oil Parameter, unit	Equipment	Recommended ASTM Method
Kinematic Viscosity @ 40°C, cSt	Viscometer	ASTM D445
Kinematic Viscosity @ 100°C, cSt	Viscometer	ASTM D445
Dynamic Viscosity @ 150°C, cP	Tapered Bearing Simulator	ASTM D4683
Soot, mass % soot in oil	TGA	ASTM E1131
TAN, mg KOH/g	Titration	ASTM D664
TBN, mg KOH/g	Titration	ASTM D4739
Viscosity Index (VI)	Calculation	ASTM D2270
Oxidation, abs/cm	FT-IR	ASTM E2412
Nitration, abs/cm	FT-IR	ASTM E2412
Water, ppm levels	Karl Fisher Titration	ASTM D6304
Fuel Dilution, mass % fuel in oil	Gas Chromatography	ASTM D7593
Wear Metal - Aluminum (Al), ppm	ICP	ASTM D5185
Wear Metal - Chromium (Cr), ppm	ICP	D5185
Wear Metal - Copper (Cu), ppm	ICP	D5185
Wear Metal - Iron (Fe), ppm	ICP	D5185
Wear Metal - Lead (Pb), ppm	ICP	D5185
Wear Metal - Tin, ppm	ICP	D5185

Contaminant Metal - Potassium (K), ppm	ICP	D5185
Contaminant Metal - Silicon (Si), ppm	ICP	D5185
Contaminant Metal - Sodium (Na), ppm	ICP	D5185
Additive Metal - Calcium (Ca), ppm	ICP	D5185
Additive Metal - Barium (Ba), ppm	ICP	D5185
Additive Metal - Magnesium (Mg), ppm	ICP	D5185
Additive Metal - Zinc (Zn), ppm	ICP	D5185
Additive Metal - Phosphorus (P), ppm	ICP	D5185
Contaminant Metal - Chlorine (Cl), ppm	Various Methods	Various Methods
Other Elements, ppm	ICP	D5185

Documentation of the lubricating oil drain interval validation **must** be retained in the site assessment.

Criteria 1: Linear trending throughout each lubricating oil drain interval of:

- Total Base Number (TBN)
- Total Acid Number (TAN)
- Oxidation
- Nitration
- Soot (There is a possibility that soot will **not** begin trending immediately; when trending, soot is usually linear.
- Iron
- Lead
- Copper

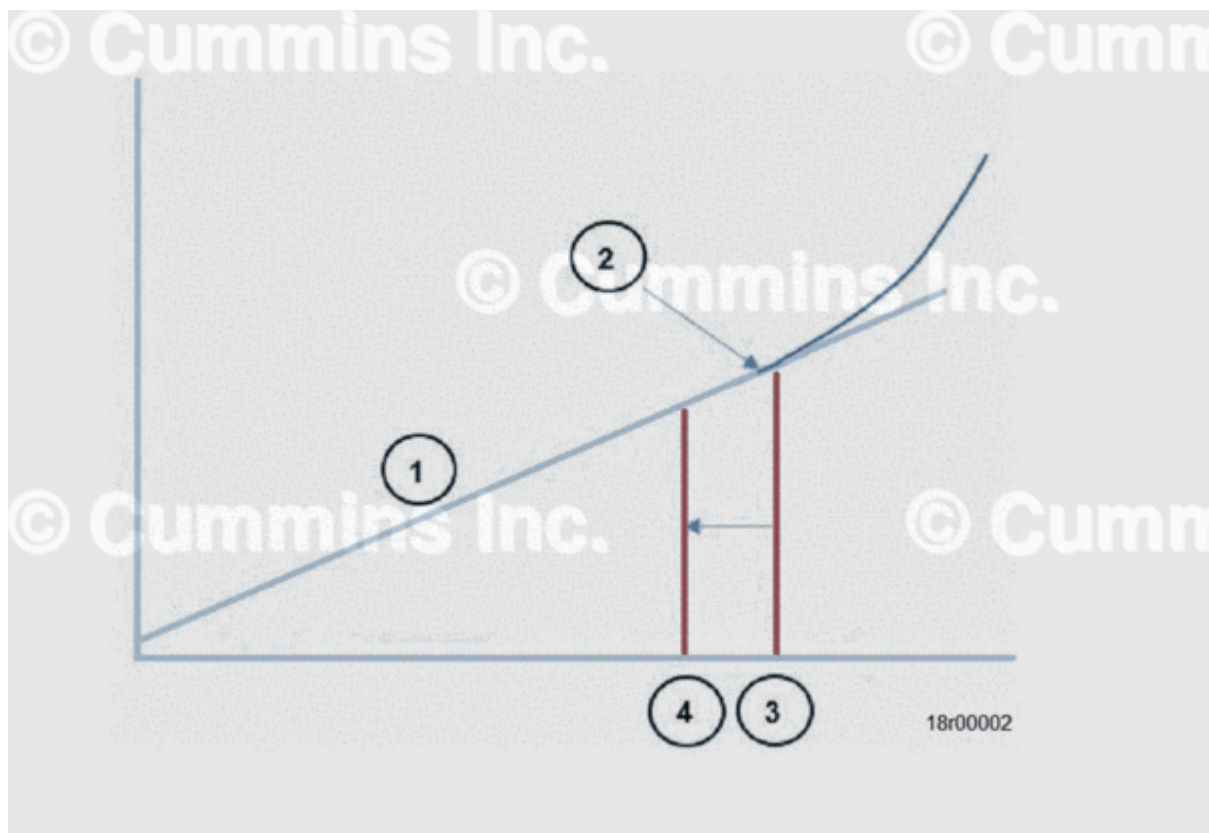


Figure 1, Linear Trending

- Linear trend of lubricating oil analysis parameter value versus hours on lubricating oil.
- Point at which lubricating oil analysis parameter value deviates from linear trend line.
- Hours on lubricating oil at which lubricating oil analysis parameter value deviates from linear trend line.
- Established lubricating oil drain limit. Subtract 20% from hours on lubricating oil where parameter value deviates from linear trend line.

Document History

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